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Calculus & Analytical Geometry

1. The training cost of an AI model is given by C(x) = $\frac{(x^8 - 256)}{(x^4 - 16)}$, where x represents the learning rate. Find the limit as x → 2.

2. The activation response of a neural network is A(x) = $\frac{(\sqrt{(x + 4)} - 2)}{x}$, where x is a very small change in signal input. Evaluate the limit as x → 0.

3. The prediction accuracy of an AI algorithm is P(x) = $\frac{(x^3 - 8)}{(x - 2)}$, where x denotes the number of data batches. Find the limit as x → 2.

4. The loss function during training is L(x) = $\frac{(x^2 + 3x - 4)}{(x + 4)}$, where x represents the number of iterations. Determine the limit as x → −4.

5. Dataset growth is modeled by G(x) = $\frac{(\sqrt{x} - 3)}{(x - 9)}$, where x is the dataset size. Find the limit as x → 9.

6. The convergence speed of an AI system is S(x) = $\frac{(x^2 - 4x + 4)}{(x - 2)}$ where x represents training epochs. Evaluate the limit as x → 2.

7. Model stability is expressed by M(x) = $\frac{(\sqrt{(x + 1)} - \sqrt{2})}{(x - 1)}$, where x denotes the noise factor. Find the limit as x → 1.

8. The optimizer response of a learning algorithm is R(x) = $\frac{(x^3 - 1)}{(x - 1)}$, where x is the step size. Evaluate the limit as x → 1.

9. Training efficiency of a model is given by E(x) = $\frac{(x^2 - 9)}{(x - 3)}$, where x represents the number of hidden layers. Find the limit as x → 3.

10. Memory utilization of an AI system is modeled by U(x) = $\frac{((4 + x)^2 - 4)}{x}$, where x is the additional memory load. Evaluate the limit as x → 0.

11. Network vulnerability is given by V(x) = $\frac{(e^{7x} - 1)}{x}$, , where x denotes firewall strength. Find the limit as x → 0.

12. Encryption delay is modeled by D(x) = $\frac{(\sqrt{(x+9)}-3)}{x}$, where x represents network lag. Evaluate the limit as x → 0.

13. System threat level is described by T(x) = $\frac{(x^3-27)}{(x-3)}$, where x indicates attack intensity. Find the limit as x → 3.

14. Authentication performance is modeled by A(x) = $\frac{(x^2+x)}{x}$,where x is the number of login attempts. Evaluate the limit as x → 0.

15. Malware detection rate is expressed as M(x) = $\frac{(\sqrt{(x+16)}-4)}{x}$, where x denotes signature updates. Find the limit as x → 0.

16. Server load is modeled by S(x) = $\frac{(x^2-1)}{(x+1)}$, where x represents the number of data packets. Evaluate the limit as x → −1.

17. Risk assessment is given by R(x) = $\frac{(x^2-25)}{(x-5)}$, where x is the risk index. Find the limit as x → 5.

18. Firewall response is described by F(x) =$\frac{(\sqrt{(x+1)}-1)}{x}$, where x is a sudden traffic spike. Evaluate the limit as x → 0.

19. Data breach probability is modeled by B(x) = $\frac{(x^3+1)}{(x+1)}$, where x denotes a vulnerability factor. Find the limit as x → −1.

20. Cyber-attack growth is given by G(x) = $\frac{(x^2-6x+9)}{(x-3)}$, where x represents attack cycles. Evaluate the limit as x → 3.

21. Image rendering time is modeled by R(x) = $(1+\frac{x}{2})^{\frac{3}{x}}$, where x represents the number of effects layers. Find the limit as x → 0.

22. Video compression efficiency is expressed as C(x) = $\frac{(\sqrt{(x+25)}-5)}{x}$, where x is the compression factor. Evaluate the limit as x → 0.

23. Animation smoothness is given by S(x) = $(1-\frac{x}{2})^{\frac{3}{x}}$, where x represents the frame rate. Find the limit as x → 0.

24. Graphic quality index is modeled by Q(x) = $(1+\frac{2}{x})^x$, where x denotes resolution parameter. Evaluate the limit as x → ∞.

25. Color contrast variation is expressed as K(x) = $\frac{(\sqrt{(x+4)}-2)}{x}$ where x is a small change in brightness. Find the limit as x → 0.

26. Poster design efficiency is modeled by E(x) = e^{-x} where x represents the number of design revisions. Evaluate the limit as x → ∞.

27. Media file size is expressed by F(x) = $\frac{(2^x - 3^x)}{x}$, where x denotes the number of layers. Find the limit as x → 0.

28. Visual rendering speed is modeled by V(x) = $\frac{(\sqrt{(x+16)}-4)}{x}$ where x is GPU load. Evaluate the limit as x → 0.

29. Advertising campaign reach is given by A(x) = $\frac{(2^{-x} - 3^{-x})}{x}$, where x represents the number of media platforms. Find the limit as x → 0.

30. User-interface responsiveness is modeled by U(x) = $\frac{(x^2 - 6x + 9)}{(x-3)}$, where x denotes user interactions. Evaluate the limit as x → 3.

31. AI System Threshold Behavior

An AI classifier changes its decision rule at a confidence score of x = 0.

The response function is:

$$f(x) = \begin{cases} x^2 + 1, & x < 0 \\ 2x + 1, & x \geq 0 \end{cases}$$

Find:

(a) Left-hand limit at x = 0

(b) Right-hand limit at x = 0

32. Cyber Security Firewall Rule Switching

A network firewall switches between two different packet-filtering strategies based on traffic load x. The firewall behavior is modeled by the function:

$$f(x) = \begin{cases} 5 - x, & x < 2 \\ x^2 - 3, & x \geq 2 \end{cases}$$

Evaluate the left-hand and right-hand limits as $x \to 2$.

33. MMG Video Rendering Mode Change

A video editor switches rendering engines at frame rate $x = 30$:

$$f(x) = \begin{cases} \sqrt{(x+6)}, & x < 30 \\ \dfrac{x}{5}, & x \geq 30 \end{cases}$$

Find the limits at $x = 30$.

34. AI Learning Rate Adjustment

An AI optimizer behaves differently for learning rates below and above $x = 1$:

$$f(x) = \begin{cases} \ln(2 - x), & x < 1 \\ x^2 - 1, & x \geq 1 \end{cases}$$

Find both one-sided limits as $x \to 1$.

35. Cyber Security Authentication Delay

Authentication delay is modeled by:

$$f(x) = \begin{cases} \dfrac{1}{x - 3}, & x < 3 \\ \sqrt{x - 3}, & x \geq 3 \end{cases}$$

Find the left-hand and right-hand limits as $x \to 3$.

36. AI Training Cost Growth

$$C(x) = \frac{(5x^2 + 3x + 1)}{(2x^2 - x + 4)}$$

Find the limit as $x \to \infty$.

37. Cyber Security Attack Frequency

$$A(x) = \frac{(7x - 10)}{\sqrt{(x^2 + 1)}}$$

Evaluate the limit as $x \to \infty$.

38. MMG Rendering Time

$$R(x) = \frac{(3x^3 - 2x + 5)}{(x^3 + 10)}$$

Find the limit as $x \to \infty$.

39. AI Model Accuracy Saturation

$$P(x) = \frac{4x^2}{(x^2 + 100)}$$

Find the limit as $x \to \infty$.

40. Cyber Security Encryption Overhead

$$E(x) = \sqrt{\frac{(x^2 + 5x)}{(x^2 + 1)}}$$

Evaluate the limit as $x \to \infty$.

41. AI Loss Function Stability

$$L(x) = \begin{cases} x^2, & x \neq 2 \\ 5, & x = 2 \end{cases}$$

Determine whether the function is continuous at $x = 2$.

42. Cyber Security Risk Index

$$R(x) = \frac{(x^2 - 4)}{(x - 2)}$$

Determine whether the function is continuous at $x = 2$.

43. MMG Color Intensity Adjustment

$$L(x) = \begin{cases} \sqrt{x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

Check the continuity of the function at $x = 0$.

44. AI Activation Function

$$A(x) = \begin{cases} x+1, & x < 1 \\ 2, & x = 1 \\ x^2, & x > 1 \end{cases}$$

Determine whether the function is continuous at x = 1.

45. Cyber Security Traffic Control

$$T(x) = |x - 3|$$

Analyze whether the function is continuous at x = 3

46. MMG UI Response Function

$$U(x) = \begin{cases} \dfrac{(x^2 - 1)}{(x - 1)}, & x \neq 1 \\ 2, & x = 1 \end{cases}$$

Determine if the function is continuous at x = 1.

47. Evaluate the following limits of the following function.

a) $\lim\limits_{x \to 0} \frac{1 - cos2x}{1 - cos3x}$
b) $\lim\limits_{x \to \frac{\pi}{2}} \frac{\tan(cosx)}{cosx}$
c) $\lim\limits_{x \to 0} \frac{cosecx - cotx}{x}$
d) $\lim\limits_{x \to 0} \frac{1 - 3^x}{1 - 2^{-x}}$

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